

SRE STUDIUM PRZYPADKU

AZURE ARC HYBRID CLOUD LAB

Projekt Proof-of-Value: hybrydowa orkiestracja chmury, observability, FinOps, IaC, Policy as Code i CI/CD pipeline.

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Firma: IT Solutions Bartosz Suszko, Olsztyn

Data: Maj 2026 (Tydzien 1-2)

GitHub: github.com/buth11/sre-homelab-azure-arc

Stack: K3s | Azure Arc | Prometheus | Grafana | Terraform | KQL | Azure Policy | Azure DevOps

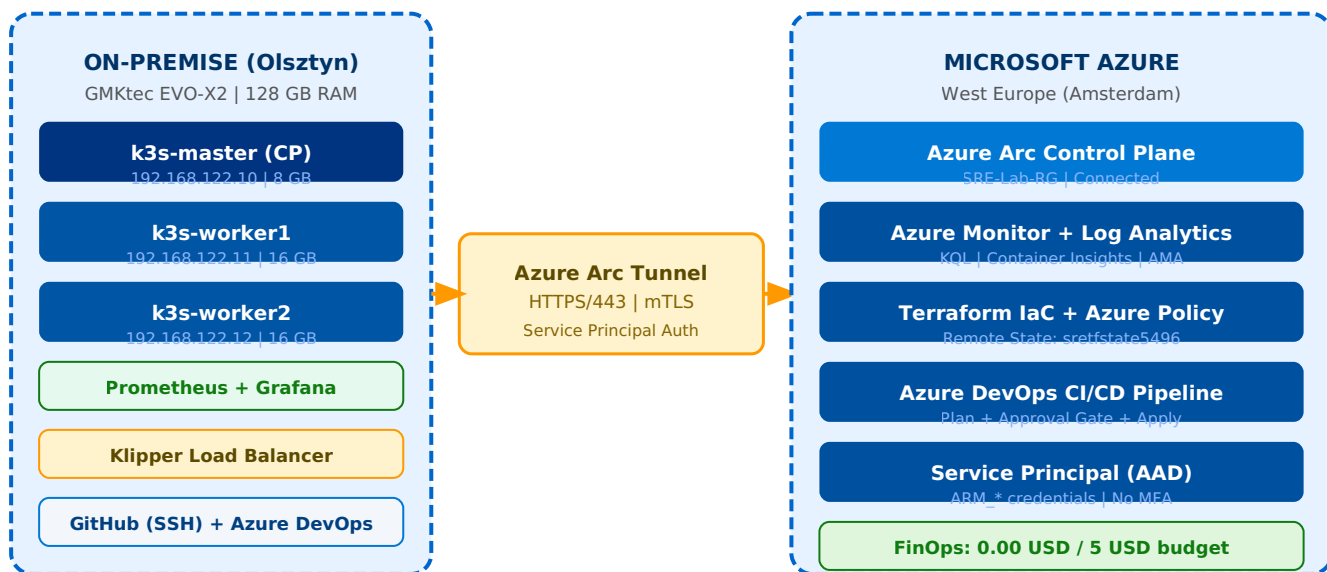
Spis treści

1. Kontekst biznesowy i architektura hybrydowa
2. Infrastruktura On-Premise — klaster K3s
3. FinOps — kontrola kosztów
4. Integracja hybrydowa — Azure Arc
5. Observability — Container Insights i Azure Monitor
6. Zapytania KQL — diagnostyka i SLO
7. Load Balancing — weryfikacja round-robin
8. Prometheus i Grafana — lokalny monitoring stack
9. Infrastructure as Code — Terraform
10. Policy as Code — Azure Policy
11. GitHub — dokumentacja i automatyzacja
12. Postmortem — POST-001 etcd Split-Brain
13. Wyniki tygodnia 1
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15. Troubleshooting Log

Rozdział 1: Kontekst i architektura

Projekt demonstruje budowę produkcyjnego środowiska hybrydowego na GMKtec EVO-X2 (128 GB RAM) łączącego K3s Kubernetes z Microsoft Azure przez Azure Arc.

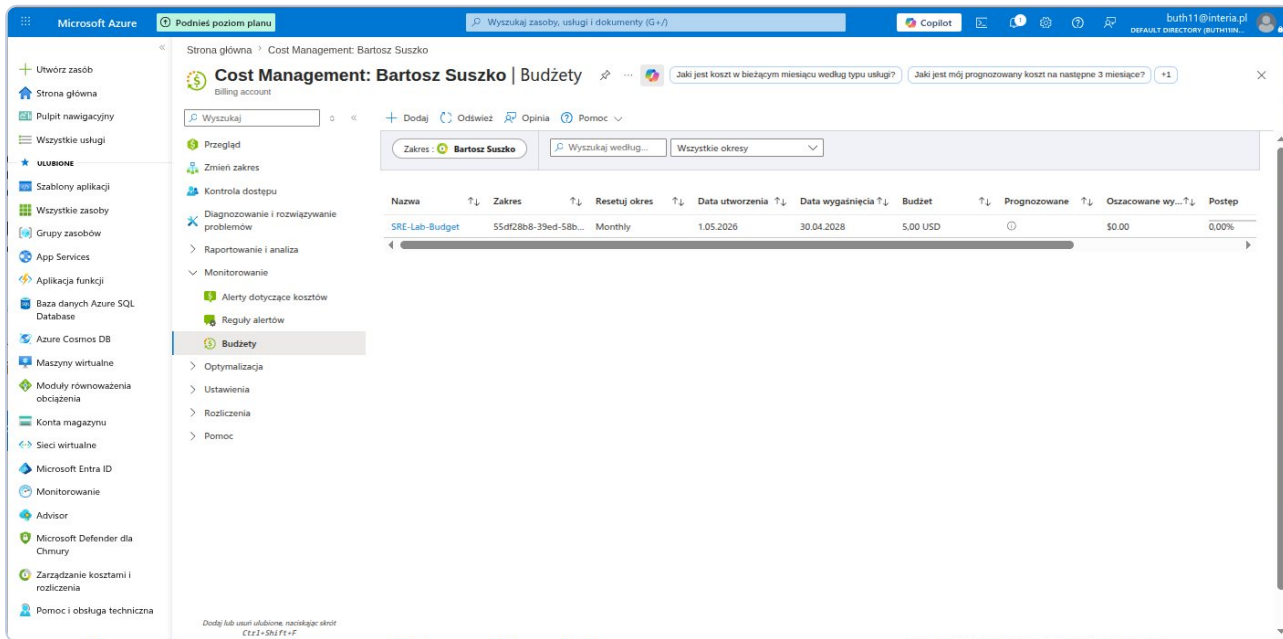
Diagram architektury (Tydzień 1+2)



Rozdział 2: Infrastruktura On-Premise

Wezel	Rola	IP	CPU	RAM
k3s-master	Control Plane	192.168.122.10	4 vCPU	8 GB
k3s-worker1	Worker	192.168.122.11	4 vCPU	16 GB
k3s-worker2	Worker	192.168.122.12	4 vCPU	16 GB

Rozdział 3: FinOps



The screenshot shows the Microsoft Azure Cost Management interface for user Bartosz Suszko. The left sidebar contains navigation links for various Azure services. The main content area is titled 'Cost Management: Bartosz Suszko | Budżety' and displays a table of budgets. The table has columns for Name, Scope, Reset period, Creation date, Expiration date, Budget, Forecasted, Estimated, and Progress. The first row shows 'SRE-Lab-Budget' with a scope of '55df28b8-39ed-58b...', a monthly reset period, a creation date of 1.05.2026, an expiration date of 30.04.2028, a budget of 5.00 USD, a forecasted value of 0.00, and a progress of 0.00%.

Nazwa	Zakres	Resetuj okres	Data utworzenia	Data wygaśnięcia	Budżet	Prognozowane	Oszacowane wy...	Postęp
SRE-Lab-Budget	55df28b8-39ed-58b...	Monthly	1.05.2026	30.04.2028	5.00 USD	0.00	\$0.00	0.00%

Zrzut 1: Azure Cost Management — budżet SRE-Lab-Budget 5 USD/mies., koszt końcowy 0.00 USD.

Rozdział 4: Azure Arc

The screenshot shows the Microsoft Azure portal interface. The main content area displays the 'K3s-Homelab' cluster details under the 'Azure Arc | Klastry Kubernetes' section. The cluster is in a 'Connected' state. The 'Właściwości' (Properties) tab is selected, showing the following information:

- Wersja agenta:** 1.33.0
- Wersja platformy Kubernetes:** 1.35.4+k3s1
- Liczba węzłów:** 3
- Liczba rdzeni:** 12

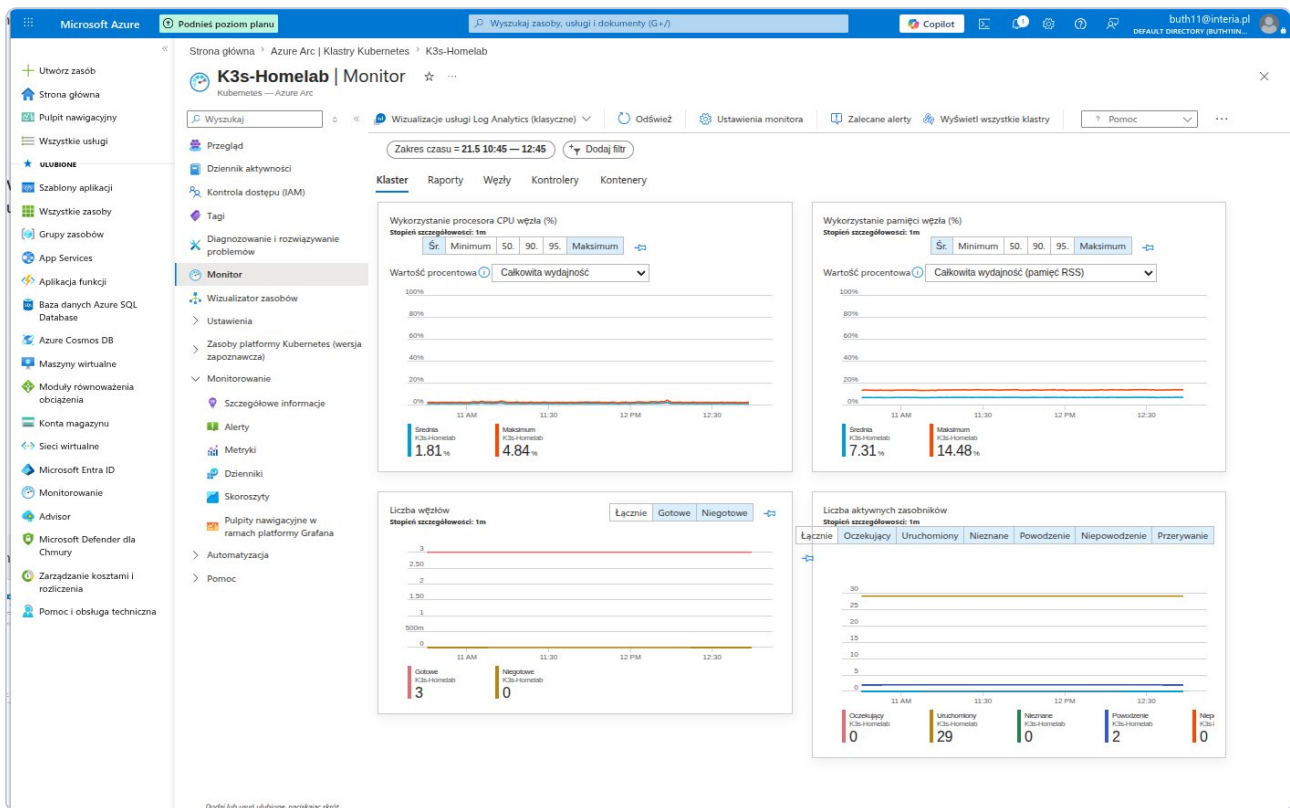
The left sidebar shows the 'Klastry Kubernetes' section expanded, and the right sidebar shows the 'Właściwości' (Properties) tab selected.

Zrzut 2: Azure Arc — K3s-Homelab Connected, v1.35.4+k3s1, 3 węzły, 12 rdzeni.

Rozdział 5: Observability

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop: bash × (buth11) 192.168.122.10 ×
{"principalId": "a3fe4a8c-7a2e-4b99-8de9-2150d617cb76",
 "tenantId": null,
 "type": "SystemAssigned"
},
"isSystemExtension": false,
"name": "azuremonitor-containers",
"packageUri": null,
"plan": null,
"provisioningState": "Succeeded",
"releaseTrain": "Stable",
"resourceGroup": "SRE-Lab-RG",
"scope": {
  "cluster": {
    "releaseNamespace": "azuremonitor-containers"
  },
  "namespace": null
},
"statuses": [],
"systemData": {
  "createdAt": "2026-05-22T07:44:41.936626+00:00",
  "createdBy": null,
  "createdByType": null,
  "lastModifiedAt": "2026-05-22T07:44:41.936626+00:00",
  "lastModifiedBy": null,
  "lastModifiedByType": null
},
"type": "Microsoft.KubernetesConfiguration/extensions",
"version": null
}
buth11@k3s-master:~$ sudo k3s kubectl get pods -n kube-system | grep ama
[sudo] password for buth11:
ama-logs-9m9x9          3/3      Running    0          37s
ama-logs-9scpr          3/3      Running    0          37s
ama-logs-rs-65b58c7f96-f8lds  2/2      Running    0          37s
ama-logs-spcdp          3/3      Running    0          37s
buth11@k3s-master:~$
```

Zrzut 3: Pody AMA Running — 4 agenty, 0 restartow, 37s po instalacji.



Zrzut 4: Azure Monitor — CPU avg 1.58%, RAM avg 6.27%, 3 węzły Ready, 29 podów.

Rozdział 6: KQL

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Dashboard', and 'All services'. The main area shows the 'Log Analytics workspace' for 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A KQL query is entered in the 'New Query 1' editor:

```
1 KubeNodeInventory
2 where TimeGenerated > ago(1h)
3 summarize arg_max(TimeGenerated, *) by Computer
4 project
5   Computer,
6   Status,
7   KubeletVersion
8   order by Computer asc
```

The query results are displayed in a table with columns: Computer, Status, and KubeletVersion. The results show 3 nodes in a 'Ready' state with version 'v1.35.4+k3s1'.

Computer	Status	KubeletVersion
k3s-master	Ready	v1.35.4+k3s1
k3s-worker1	Ready	v1.35.4+k3s1
k3s-worker2	Ready	v1.35.4+k3s1

KQL-1: 3 nodes Ready, v1.35.4+k3s1.

The screenshot shows the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Dashboard', and 'All services'. The main area displays the 'Log Analytics workspace' for 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU'. A KQL query is entered in the 'New Query' box:

```
1 KubePodInventory
2 | where TimeGenerated > ago(1h)
3 | summarize arg_max(TimeGenerated, *) by Name, Namespace
4 | summarize PodCount = count() by Namespace, PodStatus
5 | order by Namespace asc, PodCount desc
```

The results are displayed in a table with columns: Namespace, PodStatus, and PodCount.

Namespace	PodStatus	PodCount
azure-arc	Running	12
default	Running	3
kube-system	Running	14
kube-system	Succeeded	2

KQL-2: azure-arc 12 Running, default 3 Running, kube-system 14+2.

The screenshot shows the same Microsoft Azure portal interface. A new KQL query is entered in the 'New Query' box:

```
1 KubeEvents
2 | where TimeGenerated > ago(24h)
3 | project TimeGenerated, Namespace, Name, Reason, Message
4 | order by TimeGenerated desc
5 | take 20
```

The results are displayed in a table with columns: TimeGenerated [UTC], Namespace, Name, Reason, and Message.

TimeGenerated [UTC]	Namespace	Name	Reason
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-9scpr	FailedMount
5/22/2026, 7:44:56.000 AM	kube-system	ama-logs-rs-65b58c7996-f8ld5	FailedMount
5/22/2026, 7:35:32.000 AM	azure-arc	kube-aad-proxy-5f59465457-l2...	Unhealthy
5/22/2026, 7:35:25.000 AM	azure-arc	config-agent-75548f59c5-k4zrv	Unhealthy
5/22/2026, 7:35:13.000 AM	k3s-worker2		Rebooted
5/22/2026, 7:35:11.000 AM	k3s-worker1		Rebooted
5/22/2026, 7:35:09.000 AM	kube-system	metrics-server-786d997795-zz...	Unhealthy
5/22/2026, 7:35:07.000 AM	k3s-master		Rebooted
5/21/2026, 10:53:39.000 AM	kube-system	ama-logs-89qgj	Unhealthy
5/21/2026, 10:52:38.000 AM	kube-system	ama-logs-rs-65b58c7996-nlxhw	Unhealthy
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-66f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-66f7352...	FailedScheduling
5/21/2026, 8:37:39.000 AM	kube-system	svclb-aras-demo-app-66f7352...	FailedScheduling

The message for the last event is: "0/3 nodes are available: 1 node(s) didn't have free ports for the requested pod ports, 2 node(s) didn't satisfy ... s are available: 1 node(s) didn't have free ports for the requested pod ports, 2 Preemption is not helpful for sc..."

KQL-3: 13 eventow 24h — Rebooted, Unhealthy (chwilowe), FailedScheduling.

The screenshot shows the Microsoft Azure Log Analytics interface. On the left is the navigation pane with options like 'Create a resource', 'Dashboard', 'All services', and 'FAVORITES'. The main area displays the 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU' workspace. A KQL query is entered in the 'New Query 1' editor:

```
1 KubePodInventory
2 | where TimeGenerated > ago(24h)
3 | summarize arg_max(TimeGenerated, *) by Name, Namespace
4 | where PodRestartCount > 0
5 | project Namespace, Name, PodRestartCount, PodStatus
6 | order by PodRestartCount desc
```

The results table shows the following data:

Namespace	Name	PodRestartCount	PodStatus
azure-arc	extension-manager-85899588c...	3	Running
azure-arc	clusterconnect-agent-5bc68597...	3	Running
kube-system	svcib-traffic-4b7f5cb6-8b8kx	2	Running
azure-arc	extension-events-collector-6c9...	2	Running
azure-arc	clusteridentityoperator-767768...	2	Running
azure-arc	kube-aad-proxy-5f53465457-42...	2	Running
kube-system	svcib-traffic-4b7f5cb6-c2gs	2	Running
azure-arc	config-agent-75548f5c5-k4zrv	2	Running
azure-arc	cluster-metadata-operator-8db...	2	Running
kube-system	svcib-traffic-4b7f5cb6-9587	2	Running
azure-arc	metrics-agent-6d9947664-rcr44	2	Running
azure-arc	controller-manager-7cb948bdf...	2	Running
azure-arc	resource-sync-agent-54d88c5...	2	Running
azure-arc	flux-logs-agent-768c454d-9g...	1	Running
azure-arc	logcollector-8bd4468b4-4s2zw	1	Running
kube-system	ama-logs-e6g5p	1	Running
kube-system	helm-install-traffic-ggts	1	Succeeded
kube-system	svcib-aras-demo-app-8581479...	1	Running
default	aras-demo-app-f78659546-vb...	1	Running
kube-system	coredns-c4dffb5f-84kx	1	Running
kube-system	local-path-provisioner-5c4dc5d...	1	Running
kube-system	metrics-server-786d97795-zz...	1	Running
kube-system	svcib-aras-demo-app-8581479...	1	Running

KQL-4: 27 podow z restartami po restarcie VM — wszystkie Running.

The screenshot shows the Microsoft Azure Log Analytics interface. On the left is the navigation pane. The main area displays the 'DefaultWorkspace-b023e8b8-4446-4d4f-9fcf-d20a6c1fa093-WEU' workspace. A KQL query is entered in the 'New Query 1' editor:

```
1 KubeNodeInventory
2 | where TimeGenerated > ago(24h)
3 | summarize
4 |   TotalSamples = count(),
5 |   ReadySamples = countif(Status == "Ready")
6 | by Computer
7 | extend AvailabilityPct = round(toreal(ReadySamples) / toreal(TotalSamples) * 100, 2)
8 | project Computer, AvailabilityPct, ReadySamples, TotalSamples
9 | order by Computer asc
```

The results table shows the following data:

Computer	AvailabilityPct	ReadySamples	TotalSamples
k3s-master	100	184	184
k3s-worker1	100	184	184
k3s-worker2	100	184	184

KQL-5: SLO 100% — 184/184 probek Ready, error budget nienaruszony.

SLO 100% w oknie 24h. Target enterprise: 99.9% (43 min/mies.). Error budget nienaruszony nawet po pełnym restarcie VM.

Rozdział 7: Load Balancing

```
Desktop : bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × Desktop : bash × (buth11) 192.168.122.10 ×
https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

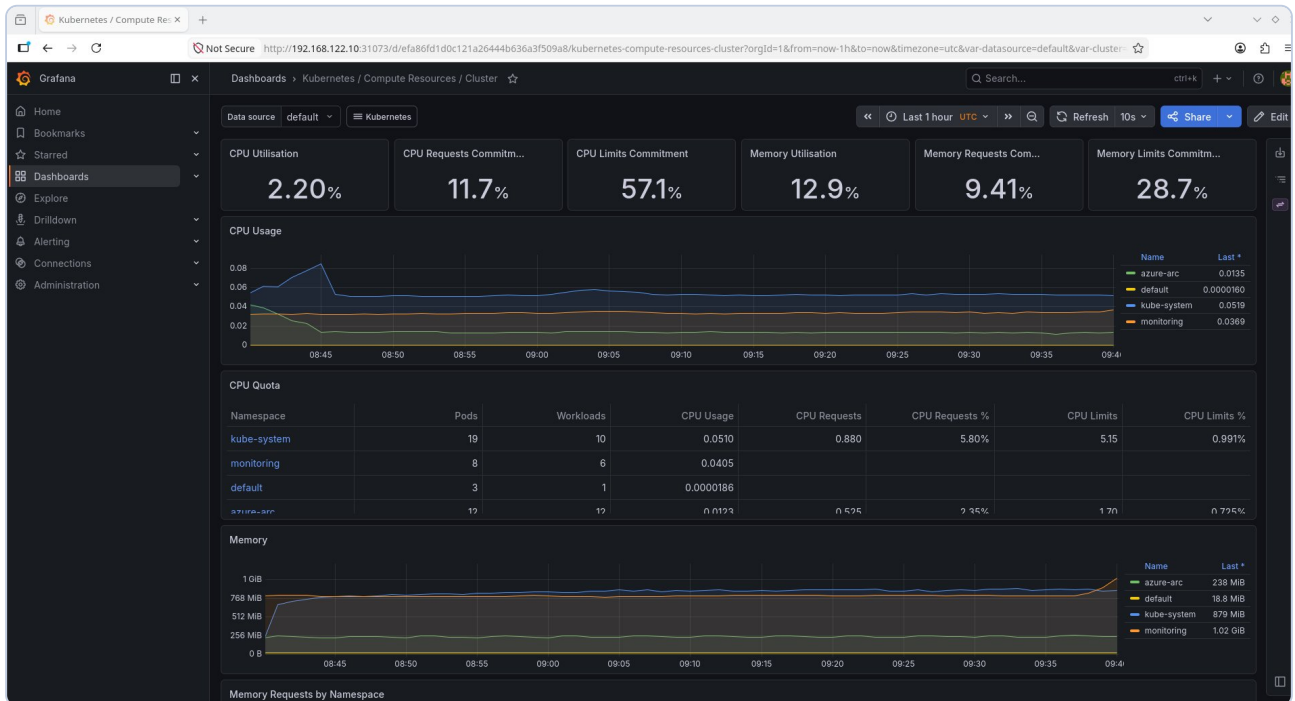
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Fri May 22 07:42:36 2026 from 192.168.122.1
buth11@k3s-master:~$ for i in {1..20}; do
  curl -s http://192.168.122.10:8080 | grep "Hostname"
  sleep 0.5
done
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-d7h9c
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-vbwkm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
Hostname: aras-demo-app-f78659546-4lprm
buth11@k3s-master:~$
```

Zrzut 5: Round-robin — 20 requestow rotujacych miedzy 3 podami, 900 rownolegla bez bledow.

Rozdział 8: Prometheus & Grafana



Zrzut 6: Grafana — CPU 2.20%, RAM 12.9%, CPU Quota per namespace.

Rozdział 9: Terraform IaC

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×
Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ action_group_id           = (known after apply)
+ budget_name               = "SRE-Lab-Budget"
+ log_analytics_workspace_id = (known after apply)
+ log_analytics_workspace_key = (sensitive value)
+ resource_group_id         = (known after apply)
+ resource_group_name        = "SRE-Lab-RG"
+ subscription_id            = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"

Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
  terraform apply "tfplan"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ terraform apply tfplan
random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=5plwbb]
azurerm_resource_group.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Creating...
azurerm_consumption_budget_subscription.sre_lab: Still creating... [00m10s elapsed]
azurerm_consumption_budget_subscription.sre_lab: Creation complete after 19s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]

Error: A resource with the ID "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG" already exists -
to be managed via Terraform this resource needs to be imported into the State. Please see the resource documentation for "azure
rm_resource_group" for more information.

with azurerm_resource_group.sre_lab,
on main.tf line 26, in resource "azurerm_resource_group" "sre_lab":
26: resource "azurerm_resource_group" "sre_lab" {

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Plan: 6 zasobow do utworzenia, 0 zmian, 0 usuniecia.

```
Desktop: bash × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × (buth11) 192.168.122.10 × terraform: bash × (buth11) 192.168.122.10 ×

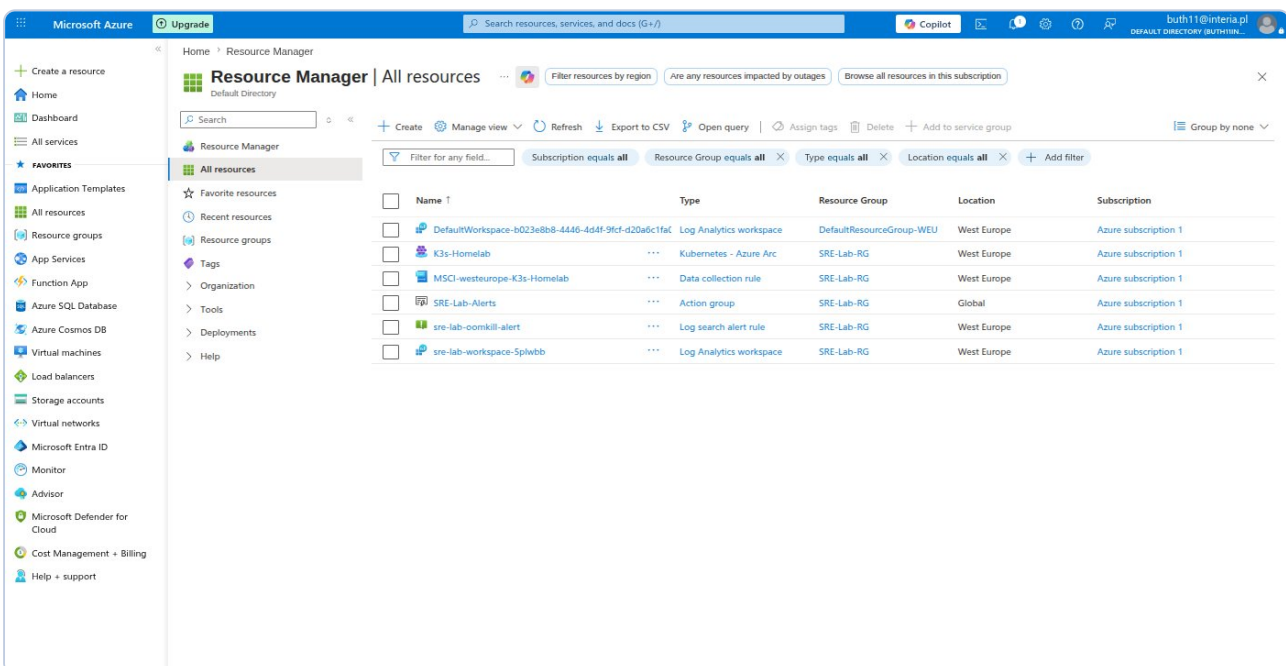
Saved the plan to: tfplan

To perform exactly these actions, run the following command to apply:
terraform apply "tfplan"
azurerm_resource_group.sre_lab: Modifying... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_resource_group.sre_lab: Modifications complete after 3s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerm_monitor_action_group.sre_alerts: Creating...
azurerm_log_analytics_workspace.sre_lab: Creating...
azurerm_monitor_action_group.sre_alerts: Creation complete after 4s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m10s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m20s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m30s elapsed]
azurerm_log_analytics_workspace.sre_lab: Still creating... [00m40s elapsed]
azurerm_log_analytics_workspace.sre_lab: Creation complete after 47s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creating...
azurerm_monitor_scheduled_query_rules_alert_v2.oom_kill: Creation complete after 5s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

Apply complete! Resources: 3 added, 1 changed, 0 destroyed.

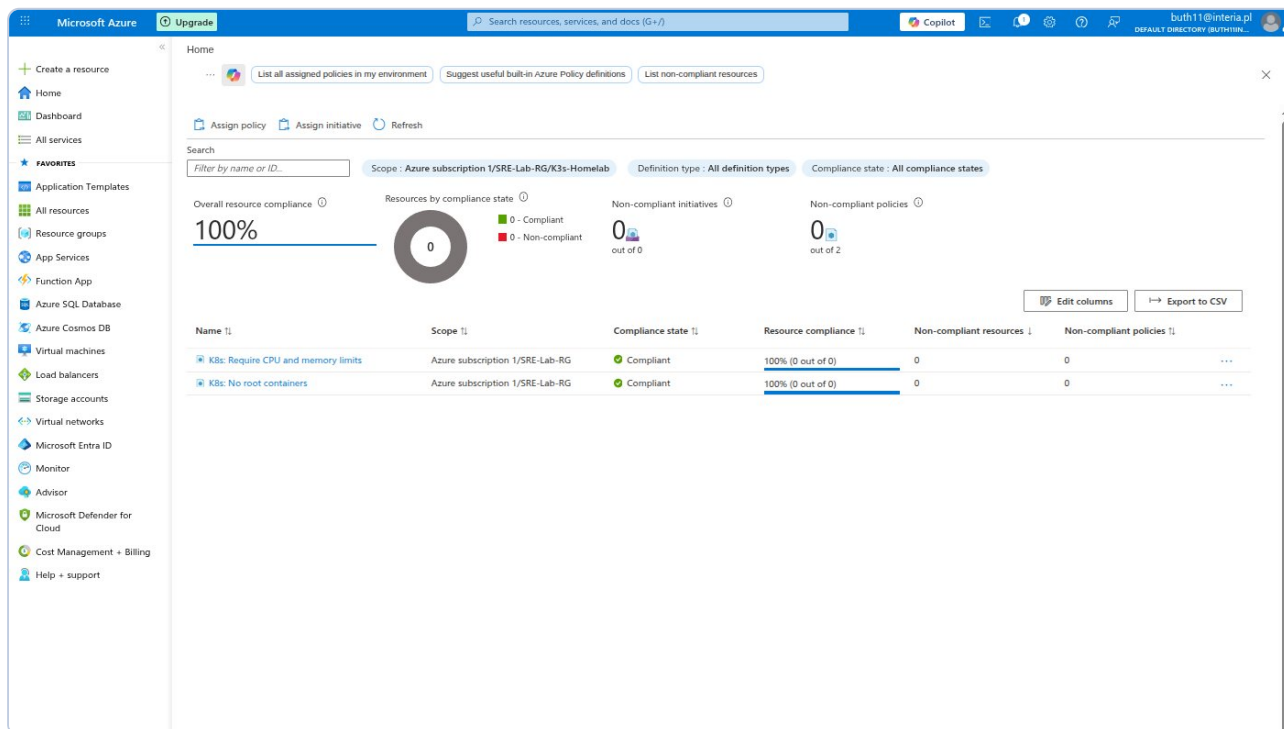
Outputs:
action_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts"
budget_name = "SRE-Lab-Budget"
log_analytics_workspace_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb"
log_analytics_workspace_key = <sensitive>
resource_group_id = "/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG"
resource_group_name = "SRE-Lab-RG"
subscription_id = "b023e8b8-4446-4d4f-9fcf-d20a6c1fa093"
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$
```

Terraform Apply: 3 dodane, 1 zmieniony, 0 usunięte.



Portal Azure: SRE-Lab-Alerts, sre-lab-oomkill-alert, sre-lab-workspace w SRE-Lab-RG.

Rozdział 10: Azure Policy



Azure Policy: K8s No root containers — Compliant, K8s CPU/memory limits — Compliant, 100%.

Rozdział 11: GitHub

The screenshot shows the GitHub repository page for 'sre-homelab-azure-arc' by user 'buth11'. The repository is public and has 1 branch (main) and 0 tags. The file list shows the initial commit 'K3s + Azure Arc SRE Homelab' with files: docs, k8s, monitoring, scripts, terraform, .gitignore, and README.md. The README content is visible, titled 'SRE Homelab: Azure Arc Hybrid Cloud Lab', describing a Proof-of-Value project. The right sidebar shows repository statistics: 0 stars, 0 forks, 0 releases, 0 packages, 1 contributor (buth11), and a language distribution bar chart showing Shell at 58.4% and HCL at 41.6%.

File	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
docs	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
k8s	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
monitoring	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
scripts	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
terraform	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
.gitignore	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago
README.md	Initial commit: K3s + Azure Arc SRE Homelab	8 minutes ago

SRE Homelab: Azure Arc Hybrid Cloud Lab

A Proof-of-Value project demonstrating hybrid cloud orchestration, infrastructure-as-code, observability, and FinOps practices using K3s Kubernetes and Microsoft Azure Arc.

Overview

Statistics:

- Stars: 0
- Forks: 0
- Releases: No releases published
- Packages: No packages published
- Contributors: 1 (buth11)
- Languages: Shell 58.4%, HCL 41.6%

GitHub repo sre-homelab-azure-arc — publiczne, Shell 58.4%, HCL 41.6%.

Rozdział 12: Postmortem POST-001

Pole	Wartosc
Severity	P2
Czas trwania	Ponizej 7 minut
Root Cause	Literowka: ks3-worker1 zamiast k3s-worker1
Status	Rozwiazany — regex validation w skryptach

Wyniki tygodnia 1

Metryka	Wartosc
Wezly klastra	3 / 3 Ready
Uruchomione pody	29
CPU utilisation	avg 2.20% / max 4.84%
RAM utilisation	avg 12.9% / max 14.48%
SLO Availability (24h)	100% — 184/184 probek Ready
Koszty Azure	0.00 USD
Zasoby Terraform	5
Polityki bezpieczenstwa	2 Compliant

Tydzien 2: Service Principal, Remote State i CI/CD Pipeline

Drugi etap projektu rozszerza infrastrukture o elementy krytyczne dla srodowisk produkcyjnych: automatyczne uwierzytelnianie bez interakcji czlowieka, centralne przechowywanie stanu Terraform z mechanizmem blokowania oraz w pelni automatyczny pipeline CI/CD z bramka zatwierdzania przed wdrozeniem.

Service Principal — automatyczne uwierzytelnianie

Azure Service Principal to konto techniczne w Azure Active Directory — odpowiednik "konta serwisowego" w systemach on-premise. Zamiast logowania przez przegladarke z MFA, pipeline uzywa identyfikatora aplikacji i hasla (client secret) zapisanych jako zmienne srodowiskowe. Konto ma role Contributor ograniczona do konkretnej subskrypcji — zasada least privilege.

```
az ad sp create-for-rbac --name "sre-homelab-sp" --role Contributor --scopes /subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093
```

Zmienne ARM_CLIENT_ID, ARM_CLIENT_SECRET, ARM_TENANT_ID, ARM_SUBSCRIPTION_ID sa automatycznie odczytywane przez Terraform i Azure CLI — eliminuje koniecznosc recznego logowania przy kazdym uruchomieniu.

Terraform Remote State w Azure Blob Storage

Lokalny plik terraform.tfstate zostal zmigrowany do Azure Blob Storage. Remote state rozwiazuje trzy problemy produkcyjne: utrate pliku przy awarii dysku, rownoczesna edycje przez wielu inzynierow (lock mechanism) i brak historii zmian (blob versioning).

Komponent	Wartosc
Storage Account	sretfstate5496 (terraform-state-rg)
Container	tfstate
Plik state	sre-homelab.tfstate (14.53 KiB)
Szyfrowanie	Server-side encryption: true
Wersjonowanie	isVersioningEnabled: true
Mechanizm blokady	Azure Blob Lease (Acquiring/Releasing state lock)

```

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # Czekaj 30 sekund na propagację uprawnień w Azure AD
sleep 30

az storage blob list \
  --container-name tfstate \
  --account-name sretfstate5496 \
  --auth-mode login \
  --query "[].{name:name, size:properties.contentLength, modified:properties.lastModified}" \
  --output table

You do not have the required permissions needed to perform this operation.
Depending on your operation, you may need to be assigned one of the following roles:
  "Storage Blob Data Owner"
  "Storage Blob Data Contributor"
  "Storage Blob Data Reader"
  "Storage Queue Data Contributor"
  "Storage Queue Data Reader"
  "Storage Table Data Contributor"
  "Storage Table Data Reader"

If you want to use the old authentication method and allow querying for the right account key, please use the "--auth-mode" parameter and "key" value.

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # Pobiera klucz dostępu do Storage Account
# auth-mode key używa klucza zamiast OAuth - omija problem z uprawnieniami SP
STORAGE_KEY=$(az storage account keys list \
  --account-name sretfstate5496 \
  --resource-group terraform-state-rg \
  --query "[0].value" -o tsv)

# Listuje pliki w kontenerze używając klucza
az storage blob list \
  --container-name tfstate \
  --account-name sretfstate5496 \
  --account-key $STORAGE_KEY \
  --query "[].{name:name, size:properties.contentLength, modified:properties.lastModified}" \
  --output table

```

Name	Size	Modified
sre-homelab.tfstate	14881	2026-05-23T18:23:07+00:00

```

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$

```

Zrzut W2-1: Terminal — plik sre-homelab.tfstate (14881 bajtów) widoczny w kontenerze tfstate po migracji ze storage account key.

The screenshot shows the Microsoft Azure portal interface. On the left is a navigation pane with options like 'Create a resource', 'Home', 'Dashboard', 'All services', and 'FAVORITES'. The main area displays the 'sre-homelab.tfstate' Blob. At the top of the main area, there are tabs for 'Overview', 'Versions', 'Snapshots', 'Edit', and 'Generate SAS'. Below these tabs is a 'Properties' section with a table of attributes. The attributes include URL, LAST MODIFIED, CREATION TIME, VERSION ID, TYPE, SIZE, ACCESS TIER, ACCESS TIER LAST MODIFIED, ARCHIVE STATUS, REHYDRATE PRIORITY, SERVER ENCRYPTED, ETAG, VERSION-LEVEL IMMUTABILITY POLICY, CACHE-CONTROL, CONTENT-TYPE, CONTENT-MD5, CONTENT-ENCODING, CONTENT-LANGUAGE, CONTENT-DISPOSITION, LEASE STATUS, LEASE STATE, LEASE DURATION, COPY STATUS, and COPY COMPLETION TIME. Below the properties table is an 'Undelete' button. At the bottom, there are sections for 'Metadata' and 'Blob index tags', each with a table for Key and Value.

Properties	
URL	https://sretfstate5496.blob...
LAST MODIFIED	5/23/2026, 8:23:07 PM
CREATION TIME	5/23/2026, 8:23:07 PM
VERSION ID	2026-05-23T18:23:07.9861361Z
TYPE	Block blob
SIZE	14.53 KiB
ACCESS TIER	Hot (Inferred)
ACCESS TIER LAST MODIFIED	N/A
ARCHIVE STATUS	-
REHYDRATE PRIORITY	-
SERVER ENCRYPTED	true
ETAG	0x8DEB8F856BD8961
VERSION-LEVEL IMMUTABILITY POLICY	Disabled
CACHE-CONTROL	
CONTENT-TYPE	application/json
CONTENT-MD5	2mHfbTY53lmCzqo/RWS+MQ...
CONTENT-ENCODING	
CONTENT-LANGUAGE	
CONTENT-DISPOSITION	
LEASE STATUS	Unlocked
LEASE STATE	Available
LEASE DURATION	-
COPY STATUS	-
COPY COMPLETION TIME	-

Metadata

Key	Value

Blob index tags

Key	Value

Zrzut W2-2: Portal Azure — wlasciwosci bloba sre-homelab.tfstate: SIZE 14.53 KiB, SERVER ENCRYPTED true, VERSION ID aktywne, LEASE STATUS Unlocked/Available.

```

buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$ # terraform plan pobierze state z Azure Blob zamiast lokalnego pliku
# Powinno pokazac "No changes" bo infrastruktura jest aktualna
terraform plan
Acquiring state lock. This may take a few moments...
random_string.suffix: Refreshing state... [id=5plwbb]
data.azurearm_subscription.current: Reading...
azurerem_resource_group.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG]
azurerem_monitor_action_group.sre_alerts: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/actionGroups/SRE-Lab-Alerts]
azurerem_log_analytics_workspace.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.OperationalInsights/workspaces/sre-lab-workspace-5plwbb]
data.azurearm_subscription.current: Read complete after 1s [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093]
azurerem_consumption_budget_subscription.sre_lab: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/providers/Microsoft.Consumption/budgets/SRE-Lab-Budget]
azurerem_monitor_scheduled_query_rules_alert_v2.oom_kill: Refreshing state... [id=/subscriptions/b023e8b8-4446-4d4f-9fcf-d20a6c1fa093/resourceGroups/SRE-Lab-RG/providers/Microsoft.Insights/scheduledQueryRules/sre-lab-oomkill-alert]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
Releasing state lock. This may take a few moments...
buth11@fedora:~/VMs/Azure/Github_repo/sre_repo/terraform$

```

Zrzut W2-3: terraform plan z remote state — "Acquiring state lock... No changes. Releasing state lock." Terraform pobiera state z Azure Blob zamiast lokalnego pliku.

Kluczowy dowód: linie "Acquiring state lock" i "Releasing state lock" w outputcie terraform plan potwierdzają że mechanizm blokowania działa. W środowisku wieloosobowym drugi inżynier nie może uruchomić apply gdy pierwszy już go wykonuje.

Azure DevOps CI/CD Pipeline

Pipeline zdefiniowany w pliku YAML (pipelines/terraform-ci.yml) automatycznie uruchamia się przy każdym pushu zmian w katalogu terraform/. Składa się z dwóch etapów z wyraźnym rozdzieleniem odpowiedzialności.

Stage	Kroki	Czas	Trigger
Terraform Validate & Plan	install, init, validate, plan, publish artifact	37s	Automatyczny przy push
Terraform Apply	init, apply (z pobranego artefaktu)	49s	Reczne zatwierdzenie (Approval Gate)

Azure DevOps | both11 / sre-homelab-arc / Pipelines / Library

Library > terraform-credentials

Variable group | Save | Clone | Security | Pipeline permissions | Approvals and checks | Help

Properties

Variable group name: terraform-credentials

Description:

☐ Link secrets from an Azure key vault as variables ⓘ

Variables

Name ↑	Value	
ARM_CLIENT_ID	54c7e017-a745-4e1e-b9da-d74f4901a3de	
ARM_CLIENT_SECRET	*****	🔒
ARM_SUBSCRIPTION_ID	b023e8b8-4446-4d4f-9fcf-d20a6c1fa093	
ARM_TENANT_ID	fd7288c4-e003-40fd-b3cf-9c0d1821cf5d	

+ Add

Zrzut W2-4: Azure DevOps Library — Variable Group "terraform-credentials" z 4 zmiennymi ARM_*. ARM_CLIENT_SECRET ukryty (ikona kluczyka) — sekrety nie są widoczne w logach pipeline.

Azure DevOps

buth11 / sre-homelab-arc / Pipelines / Environments / production

Q Search

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Automatically created environment

Add resource

Deployments Approvals and checks

Check execution order

Checks will run by order, which is pre-determined by check types. Checks within the same order will run in parallel.

[Learn more](#)

+ Add new

Order	Display name	Type	Timeout	Details
1	All approvers must approve	Approvals	30d	BS

Zrzut W2-5: Environment "production" — Approval Gate: "All approvers must approve", Timeout 30 dni. Pipeline zatrzymuje sie przed apply i wysyla powiadomienie do inzyniera.

Bartosz Suszko | SRE Case Study | 26 / 32

Azure DevOps Interface showing the 'Jobs in run' section for the pipeline 'buth11.sre-homelab-azure-arc' with ID #20260523.1. The jobs listed are:

Job Name	Status	Duration
terraform plan	Success	34s
Initialize job	Success	<1s
Checkout buth1...	Success	2s
Install Terraform...	Success	2s
terraform init	Success	7s
terraform validate	Success	1s
terraform plan	Success	15s
Publish terrafor...	Success	3s
Post-job: Chec...	Success	<1s
Finalize Job	Success	<1s

The 'Terraform Apply' section shows the job 'terraform apply' is currently pending.

The 'terraform plan' job log shows the following steps:

```
1 Agent: Azure Pipelines 1
2 Started: Just now
3 Duration: 34s
4
5 Job preparation parameters
44 1 artifact produced
45 Job live console data:
46 Starting: terraform plan
47 Async Command Start: DetectDockerContainer
48 Async Command End: DetectDockerContainer
49 Async Command Start: DetectDockerContainer
50 Async Command End: DetectDockerContainer
51 Finishing: terraform plan
```

Zrzut W2-6: Stage 1 zakończona — wszystkie kroki zielone: Install Terraform 1.15.4 (2s), terraform init (7s), terraform validate (1s), terraform plan (15s), Publish artifact (3s). 1 artifact produced.

Azure DevOps | **sre-homelab-arc** | 20260523.1

#20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform

Summary | Code Coverage

Manually run by **Bartosz Suszko** | View change | Parameters

Today at 9:13 PM | 1m 45s

Repository and version | **buth11/sre-homelab-azure-arc** | main | e9da4bfb

Related | 0 work items | 1 published

Tests and coverage | Get started

This pipeline needs permission to access a resource before this run can continue to Terraform Apply | View

Stages | Jobs

Terraform Validate ... | 1 job completed | 37s | 1 artifact

Terraform Apply | Waiting | Permission needed

Project settings

Zrzut W2-7: Pipeline oczekuje na zatwierdzenie — Stage 1 zielony (37s, 1 artifact), Stage 2 "Waiting" z komunikatem "1 approval needs your review". Widoczny przycisk Approve/Reject.

The screenshot displays the Azure DevOps web interface for a project named 'sre-homelab-arc'. The left sidebar shows the navigation menu with 'Pipelines' selected. The main area shows a specific pipeline run with the title '#20260523.1 • feat: add Azure DevOps CI/CD pipeline for Terraform'. A green checkmark indicates the run is successful. Below the title, it states 'Manually run by BS Bartosz Suszko' and 'Today at 9:13 PM' with a duration of '4m 24s'. The 'Summary' tab is active, showing details about the repository ('buth11/sre-homelab-azure-arc', main branch, commit e9da4bfb) and related items (0 work items, 1 published). The 'Stages' section shows two jobs: 'Terraform Validate ...' (1 job completed, 37s, 1 artifact) and 'Terraform Apply' (1 job completed, 49s, 1 checks passed). A 'Run new' button is visible in the top right corner.

Zrzut W2-8: Pipeline zakończona sukcesem — Terraform Validate & Plan (37s, 1 artifact) i Terraform Apply (49s, 1 checks passed) oba zielone. Całkowity czas: 4m 24s.

Na rozmowie: "Kazdy push do terraform/ uruchamia automatyczny plan. Apply wymaga ręcznego zatwierdzenia inżyniera po review planu — dokładnie tak jak w środowisku enterprise gdzie zmiana infrastruktury musi być zatwierdzona przed wdrożeniem na produkcję."

Troubleshooting Log

ISSUE-001 KQL: Failed to resolve scalar expression CPUCapacityNanoCores

project operator: Failed to resolve scalar expression named
CPUCapacityNanoCores

Przyczyna: AMA v3.x używa innych nazw kolumn niż dokumentacja dla starszego omsagent. Kolumna nie istnieje w tym schemacie.

Rozwiązanie: Uruchomiono KubeNodeInventory | getschema i uproszczono zapytanie do dostępnych kolumn.

ISSUE-002 Azure Policy: MissingPolicyParameter cpuLimit, memoryLimit

(MissingPolicyParameter) policy assignment k8s-resource-limits is missing
cpuLimit, memoryLimit

Przyczyna: Wersja 9.3.0 definicji polityki wprowadziła obowiązkowe parametry niewymagane w poprzednich wersjach.

Rozwiązanie: Dodano --params z cpuLimit: 500m i memoryLimit: 512Mi do komendy CLI.

ISSUE-003 Terraform: Resource Group already exists (brownfield import)

Error: A resource with the ID .../SRE-Lab-RG already exists – needs to be imported into State

Przyczyna: Resource Group została wcześniej stworzona ręcznie przez az connectedk8s connect. Terraform nie miał jej w state.

Rozwiązanie: terraform import azurerm_resource_group.sre_lab /subscriptions/.../SRE-Lab-RG

ISSUE-004 Terraform: Saved plan is stale after import

Error: Saved plan is stale – state was changed by another operation after plan was created

Rozwiązanie: terraform plan -out=tfplan i terraform apply tfplan po imporcie.

ISSUE-005 Container Insights: Extension missing after VM restart

Przyczyna: Certyfikat tożsamości klastra został zregenerowany przy restarcie — agenty AMA wymagały ponownej rejestracji.

Rozwiązanie: az k8s-extension create --name azuremonitor-containers — agenty wróciły do Running w 37 sekund.

ISSUE-006 POST-001: K3s etcd Split-Brain (hostname typo: ks3 vs k3s)

```
FATA[0034] Node password rejected, duplicate hostname
```

Przyczyna: Literówka ks3-worker1 zamiast k3s-worker1 przy provisioningu VM. etcd traktuje tożsamość węzła jako niemutowalną.

Rozwiązanie: `kubectl delete node + rm /etc/rancher/node/ + restart k3s-agent`. Czas resolucji: 7 minut. Skrypty zaktualizowane o regex validation.

ISSUE-007 Azure DevOps: TerraformInstaller task not found

```
A task is missing: TerraformInstaller. This task isn't installed and may be available from Marketplace.
```

Przyczyna: Rozszerzenie ms-devlabs.custom-terraform-tasks nie jest domyślnie zainstalowane w organizacji Azure DevOps.

Rozwiązanie: Instalacja z Marketplace: marketplace.visualstudio.com/items?itemName=ms-devlabs.custom-terraform-tasks — Get it free, wybór organizacji both11, Install.

ISSUE-008 Azure DevOps: Pipeline needs permission to access Variable Group

```
This pipeline needs permission to access a resource before this run can continue to Terraform Validate & Plan
```

Przyczyna: Pierwsze uruchomienie pipeline wymaga jednorazowej autoryzacji dostępu do Variable Group i Environment production.

Rozwiązanie: Kliknięcie View > Permit w oknie autoryzacji. Jednorazowe — wszystkie przyszłe uruchomienia nie wymagają ponownej autoryzacji.